Adaptive Privacy-Preserving Framework for Network Traffic Anomaly Detection

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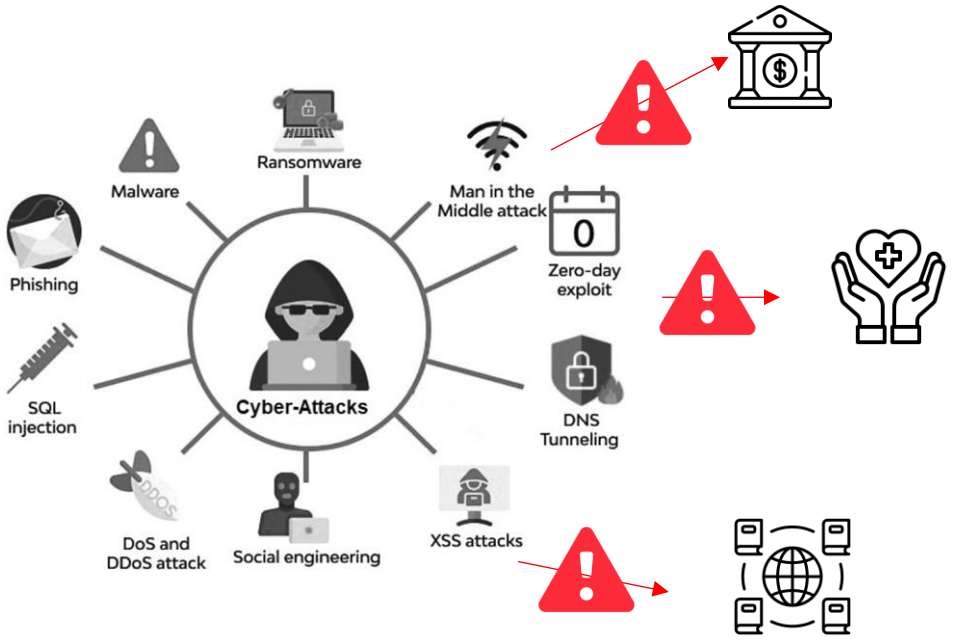
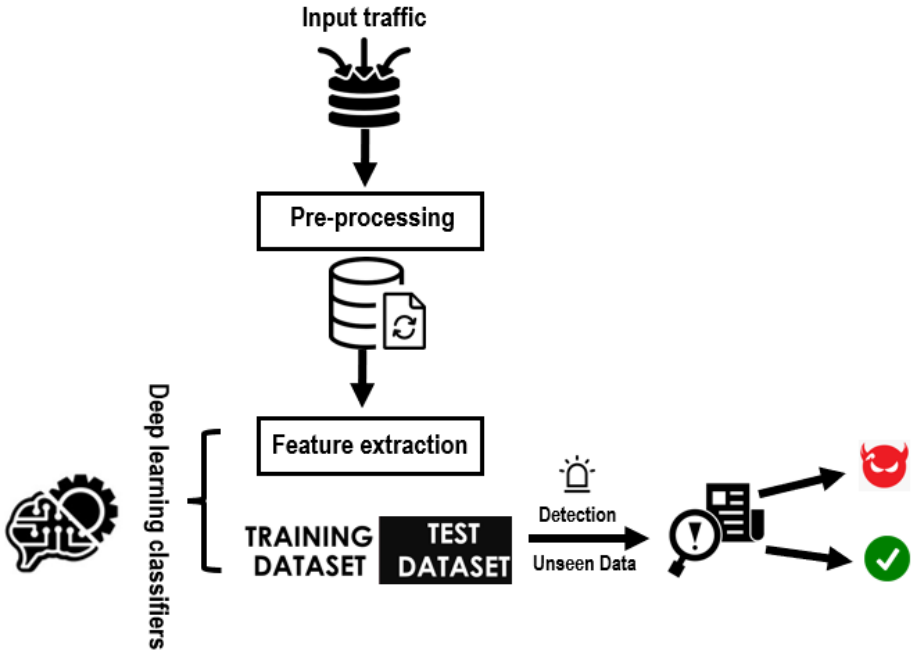
01 Background

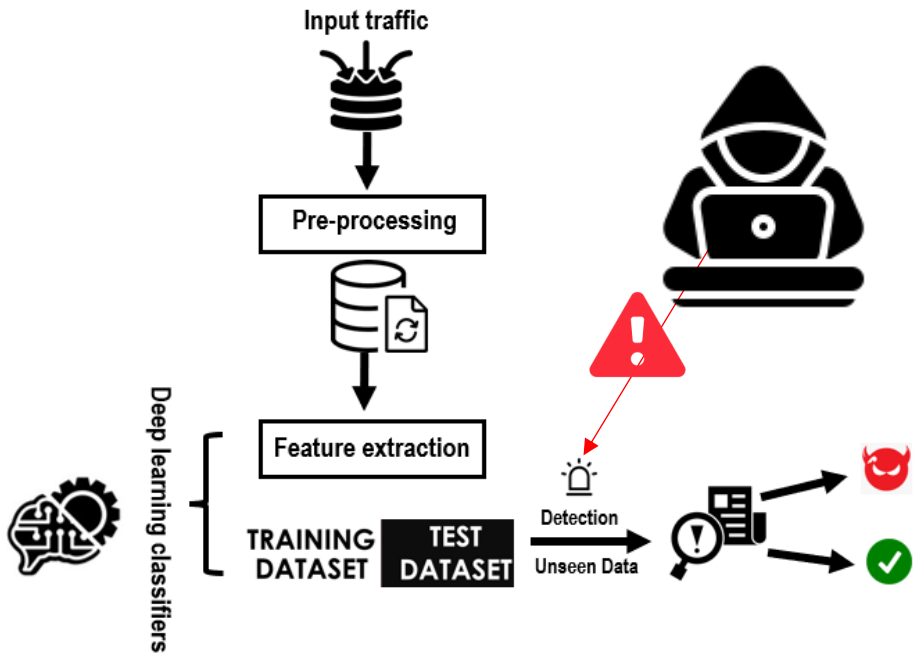
➢ **Cyberattack techniques have become increasingly sophisticated, highlighting the importance of anomaly detection systems for detecting these attacks.**





01 Challenge 1

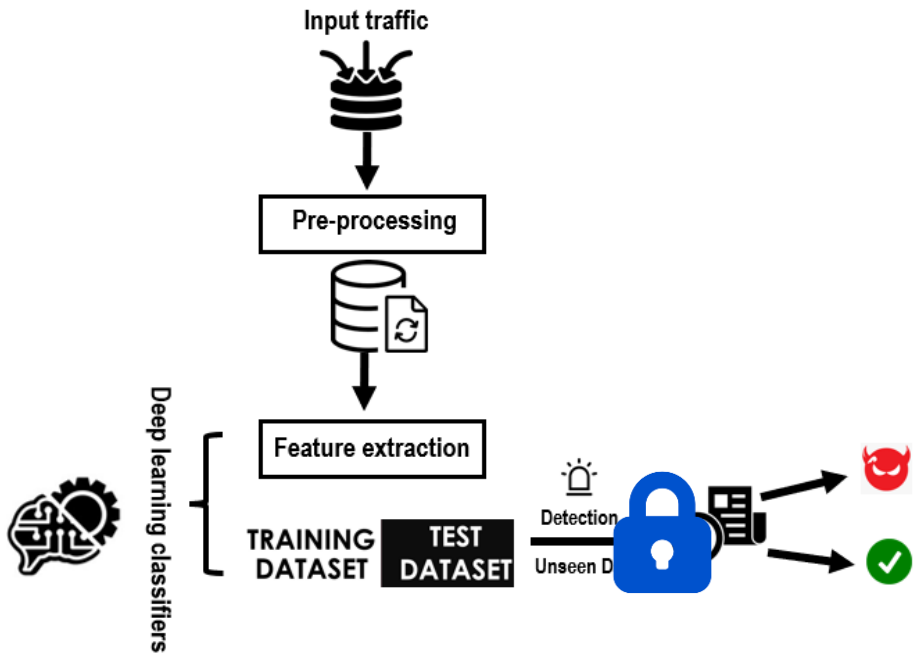
- Privacy issues in network traffic anomaly detection



- Conventional anomaly-detection systems have been designed based on the assumption that servers are trustworthy.

01 Challenge 2

- Latency and accuracy issues in privacy-preserving anomaly detection.



- The process of calculating encrypted data requires a considerably larger amount of computation than general operations, which increases latency.
- Furthermore, it reduces detection accuracy because of the noise generated during the complex calculation process.

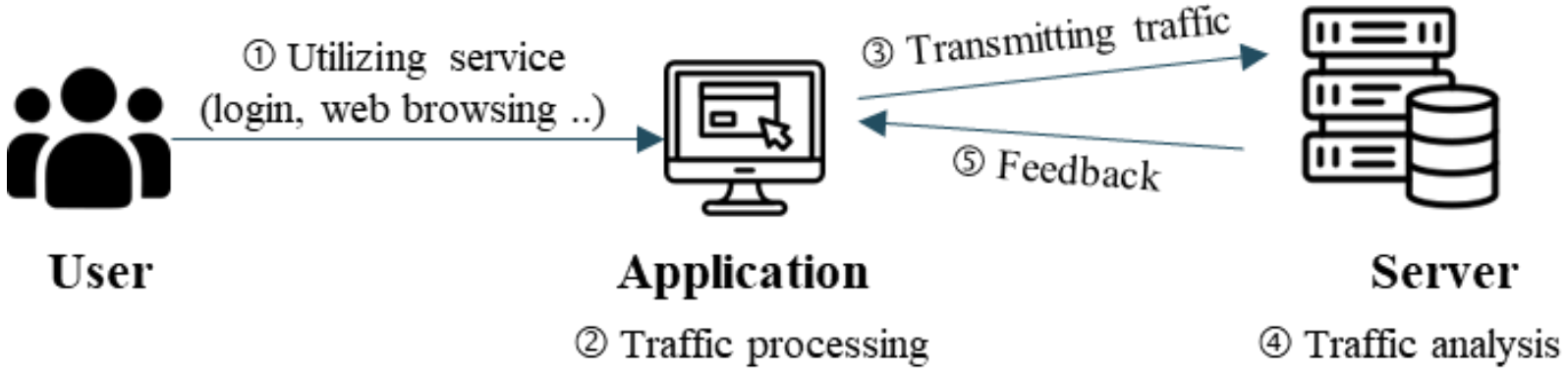
02 Goals

- **To enhance privacy,** machine learning inference with homomorphic encryption is applied during anomaly-detection tasks.
- **To reduce latency,** plaintext-based machine learning is used for anomaly-detection
- **To balance privacy and latency,** homomorphic encryption should not always be enforced; instead, the system must adaptively switch between encrypted and plaintext inference.

➢ <u>In this work, we propose a model that evaluates the privacy sensitivity of incoming data and performs anomaly detection with homomorphic encryption only for sensitive data.</u>

02 Adaptive Privacy-Preserving Anomaly Detection

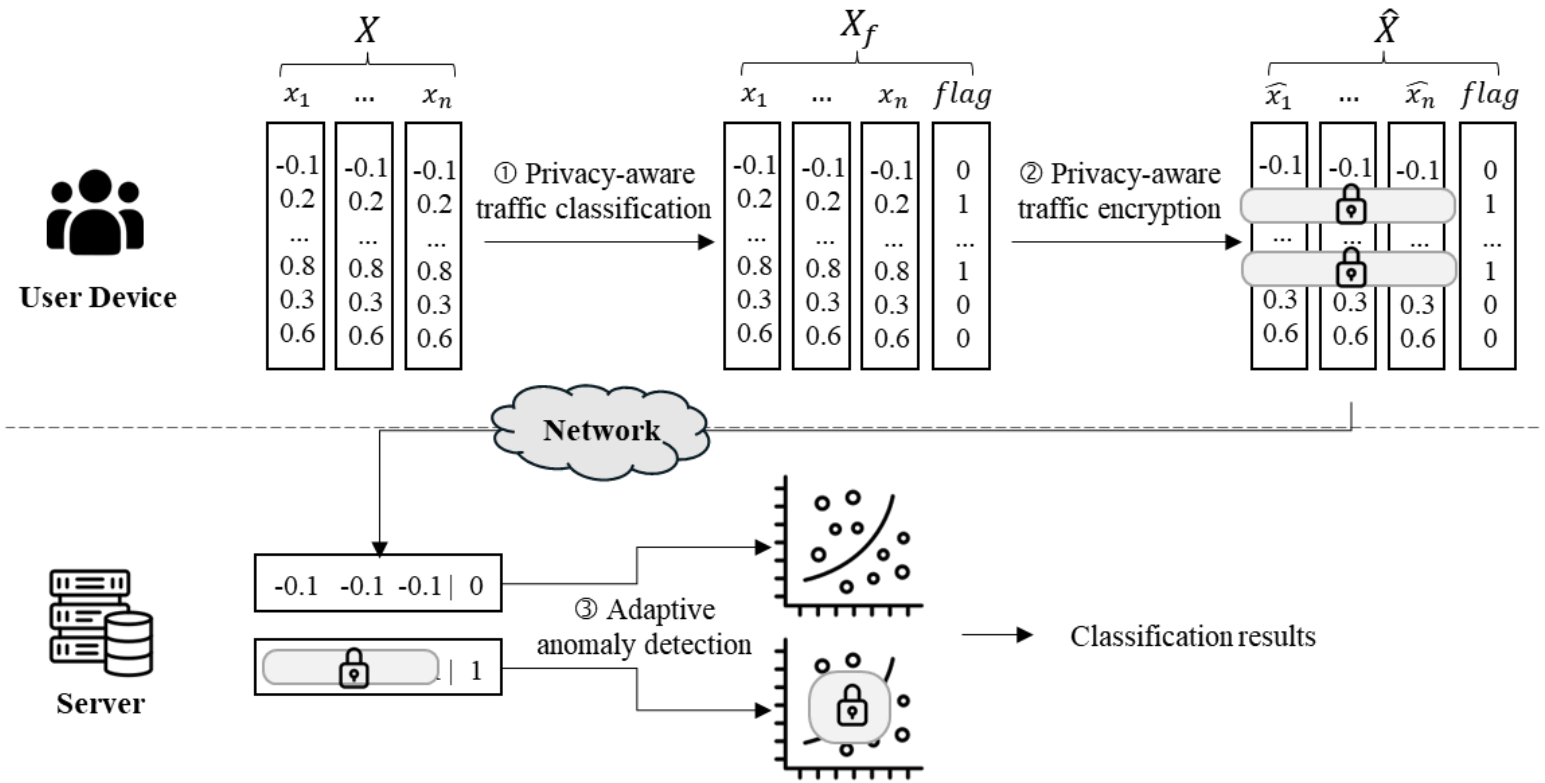
- When **a user** uses a service, traffic is generated during various behaviors such as searching, surfing, logging in.
- **The server** uses real-time traffic generated from user's behavior as input for anomaly detection.



<The scenario of the proposed APPAD>

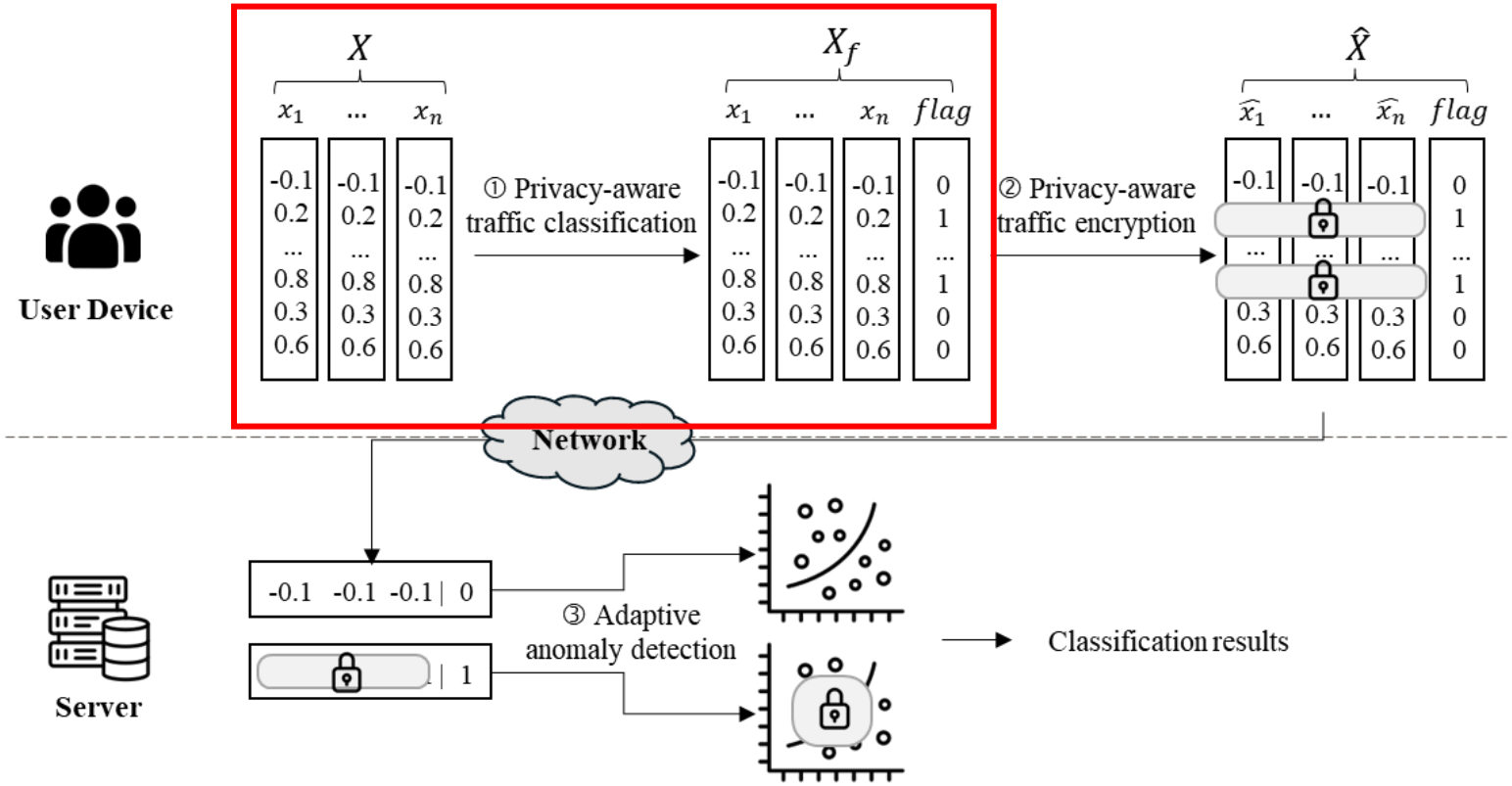
02 Adaptive Privacy-Preserving Anomaly Detection

➢ Our proposed APPAD operates in the following three steps.



<The architecture of the proposed APPAD>

02 Adaptive Privacy-Preserving Anomaly Detection

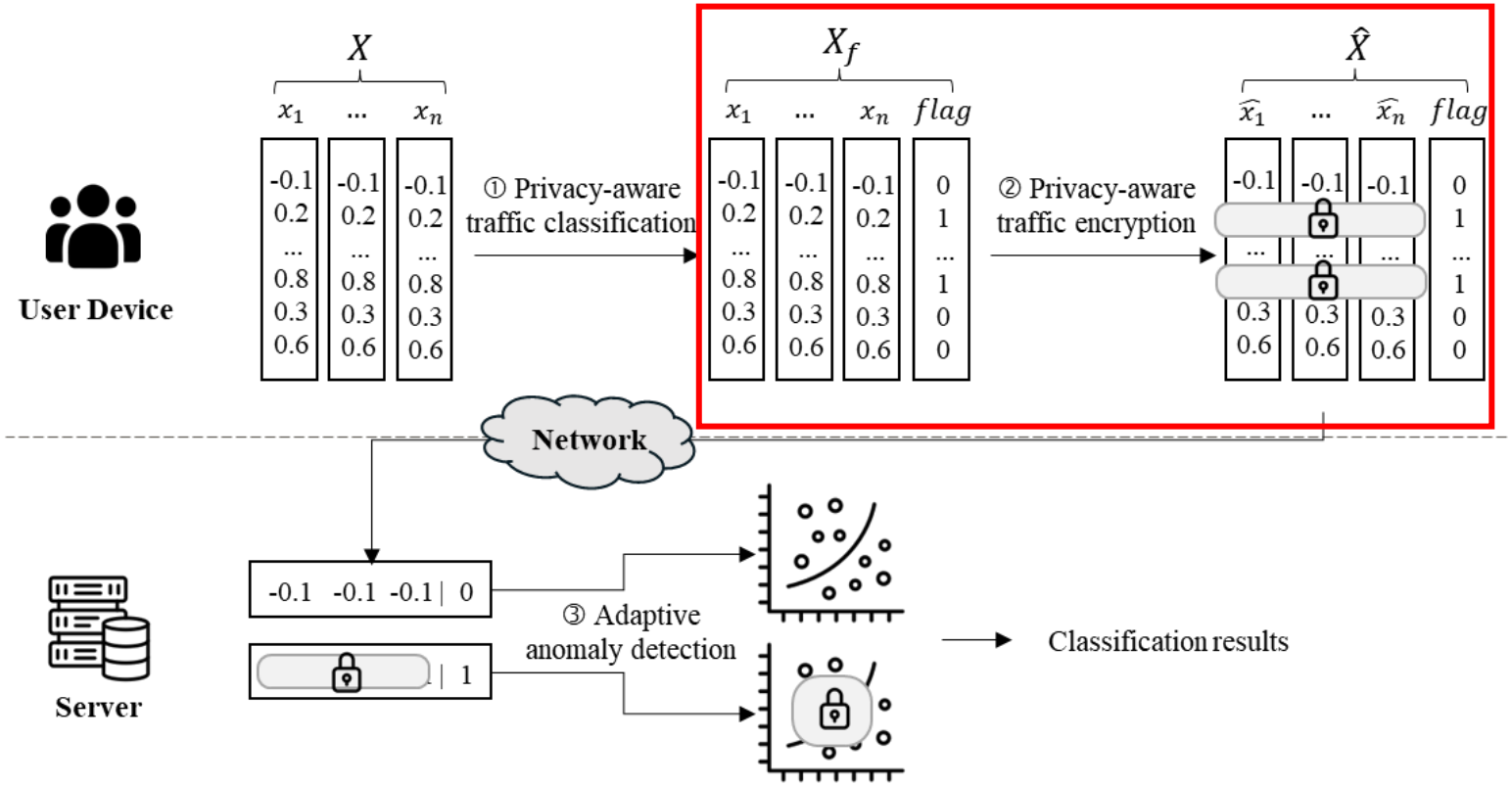


<The architecture of the proposed APPAD>

① Privacy-aware traffic classification

- **If the traffic contains private information, it is classified as privacy-sensitive traffic.**
- **If it does not contain private information, it is classified as non-sensitive traffic.**

02 Adaptive Privacy-Preserving Anomaly Detection

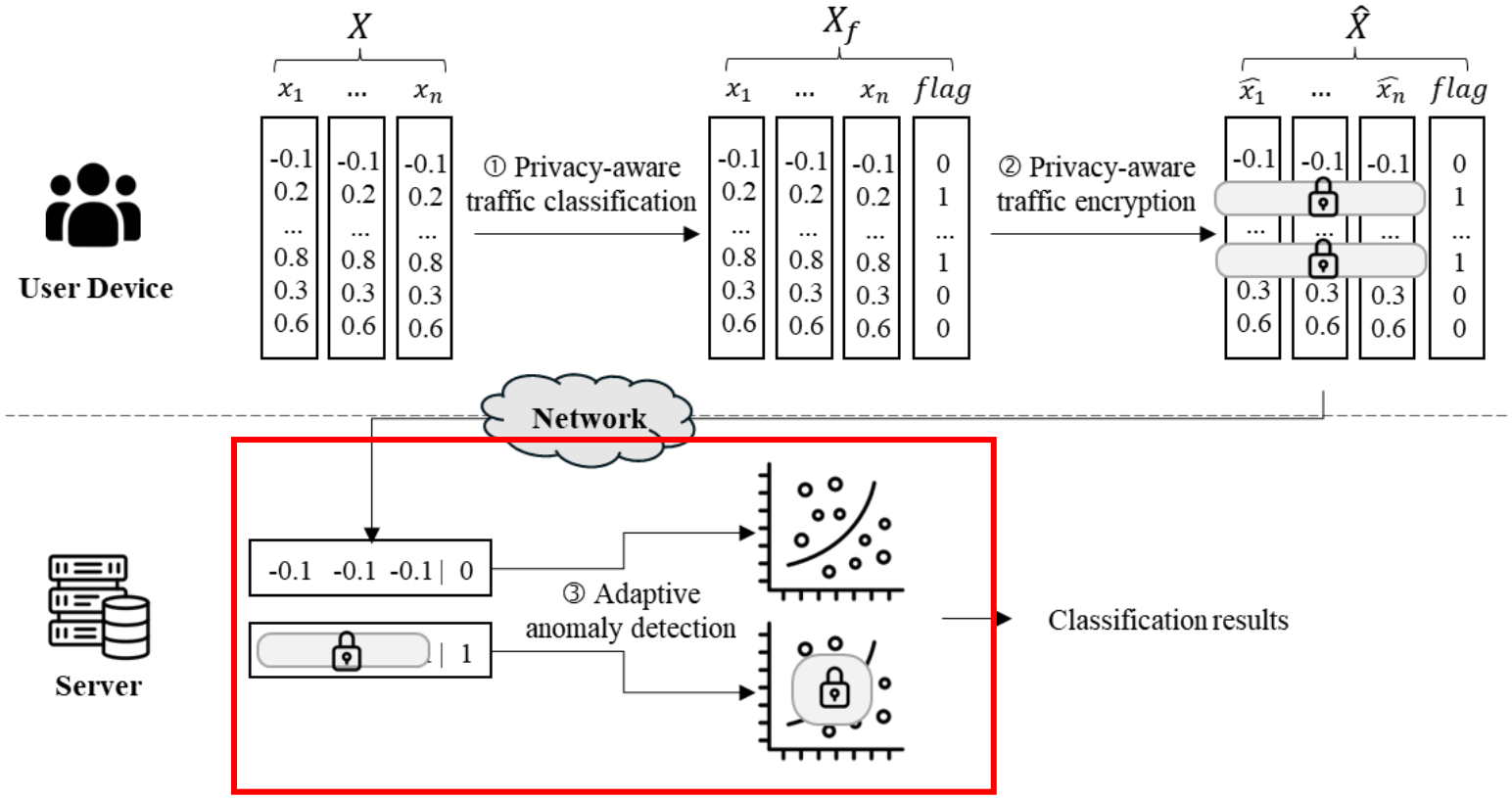


<The architecture of the proposed APPAD>

② Privacy-aware traffic encryption

- When plaintext traffic is received as input, plaintext computation is used to detect anomalies.
- When homomorphic encrypted traffic is input into an anomaly detection model, the pre-trained model outputs the encrypted results through homomorphic operations .

02 Adaptive Privacy-Preserving Anomaly Detection



<The architecture of the proposed APPAD>

③ Adaptive anomaly detection

- A logistic regression model was used as the learning and classification model for traffic anomaly detection.
- The server inspects incoming traffic and determines whether it is sensitive or non-sensitive based on the traffic's flag value.
- Homomorphic operation is used to detect anomalies when privacy-sensitive traffic is received as input.
- When non-sensitive traffic is received as input, plaintext operation is used to detect anomalies.

03 Evaluation and analysis

- **Evaluation metrics**
  - Accuracy (Acc)
  - Latency (d)
  - Information leakage rate (ILR)
  - Efficiency (M_{eff})

$$M_{eff} = \overline{Acc} \cdot (1 - \bar{d}) \cdot (1 - \overline{ILR}) \cdot \overline{SPR}$$

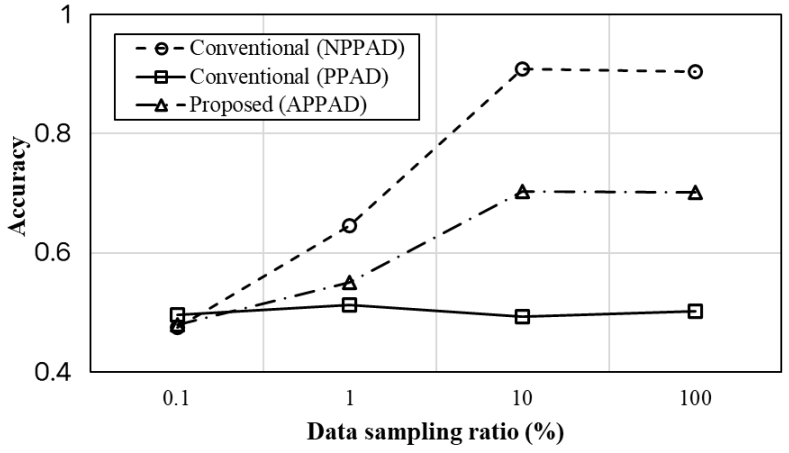
*SPR- Sensitive packet processing rate

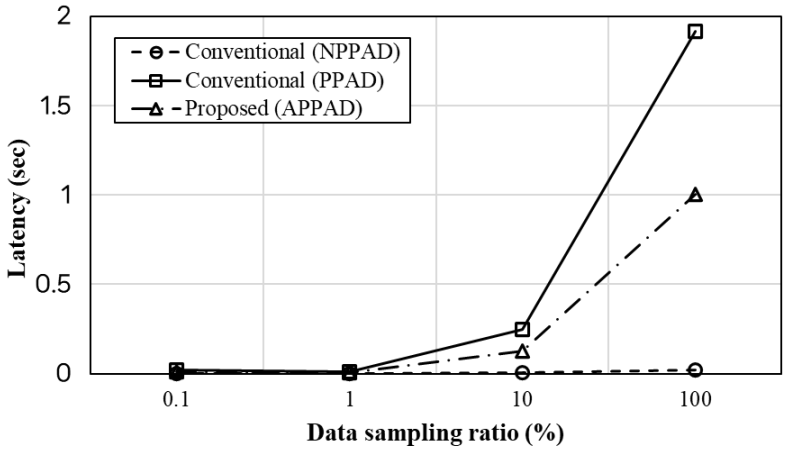
- **Comparison of Models**
  - NPPAD: Non privacy-preserving anomaly detection
  - PPAD: Privacy-preserving anomaly detection
  - APPAD: Adaptive privacy-preserving anomaly detection

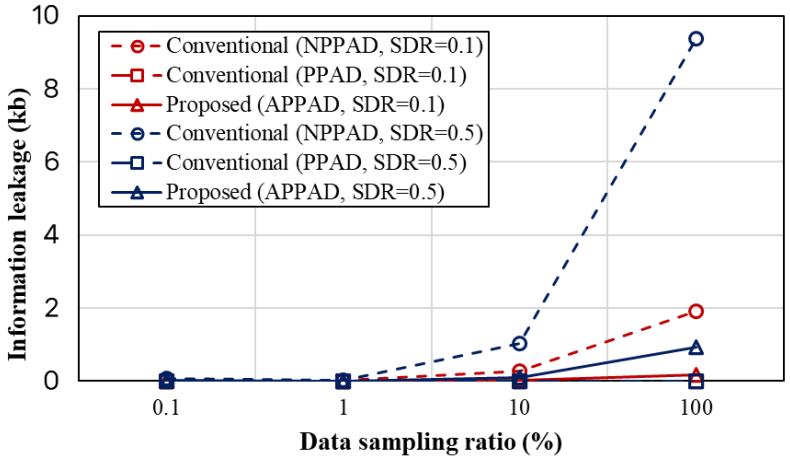
03 Evaluation and analysis

- **NPPAD: Non privacy-preserving anomaly detection**
- **PPAD: Privacy-preserving anomaly detection**
- **APPAD: Adaptive privacy-preserving anomaly detection**

- Accuracy, latency, information leakage with varying data sampling ratio.







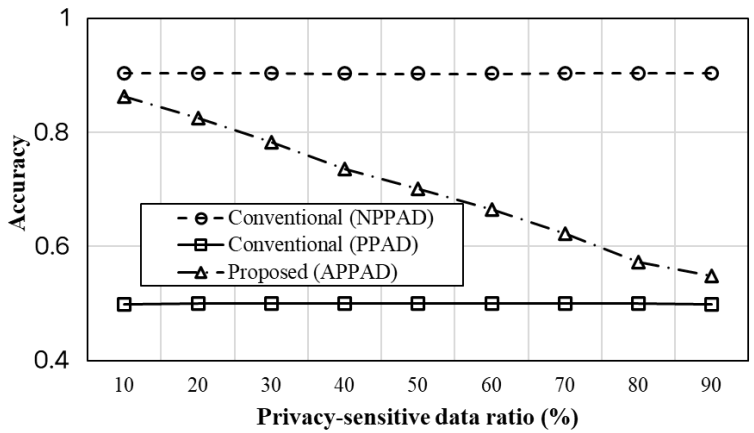
Results

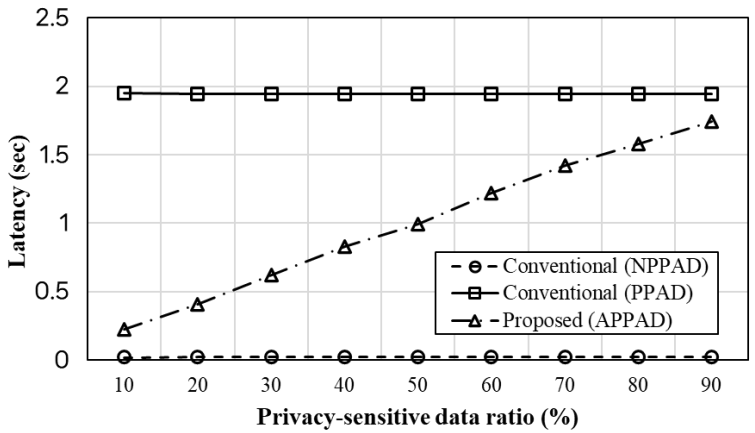
- **Accuracy: NPPAD > APPAD > PPAD**
- **Latency : PPAD > APPAD > NPPAD**
- **Information leakage : NPPAD > APPAD > PPAD**

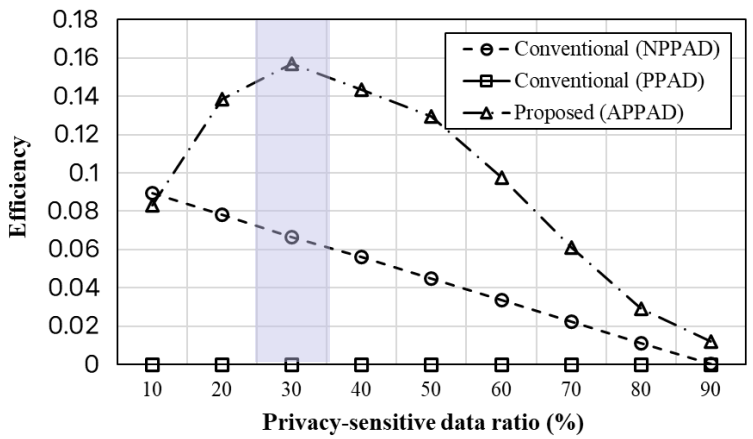
03 Evaluation and analysis

- NPPAD: Non privacy-preserving anomaly detection
- PPAD: Privacy-preserving anomaly detection
- APPAD: Adaptive privacy-preserving anomaly detection

- Accuracy, latency, efficiency with varying the ratio of privacy-sensitive data.







Results

- **Accuracy: NPPAD > APPAD > PPAD**
- **Latency: PPAD > APPAD > NPPAD**
- **Efficiency: APPAD > NPPAD > PPAD**

Conclusion

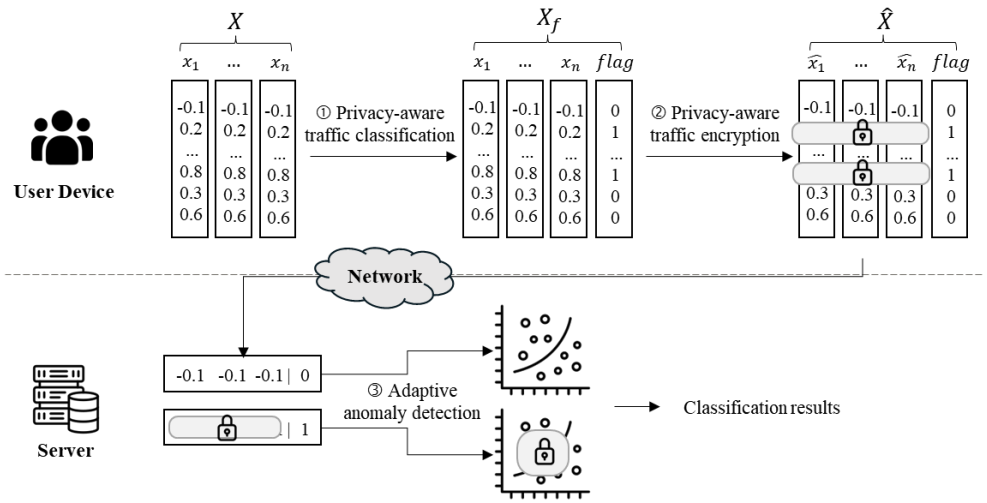
04 Summary

Challenges in anomaly detection

- Privacy issues in traditional anomaly detection.
- Performance degradation in privacy-preserving anomaly detection.

Proposed method

- This study proposes an anomaly detection method that dynamically applies homomorphic encryption based on the privacy sensitivity of data.
- The APPAD model processes incoming traffic as plaintext or ciphertext depending on the sensitivity of the traffic and performs anomaly detection through homomorphic encryption only on encrypted traffic.



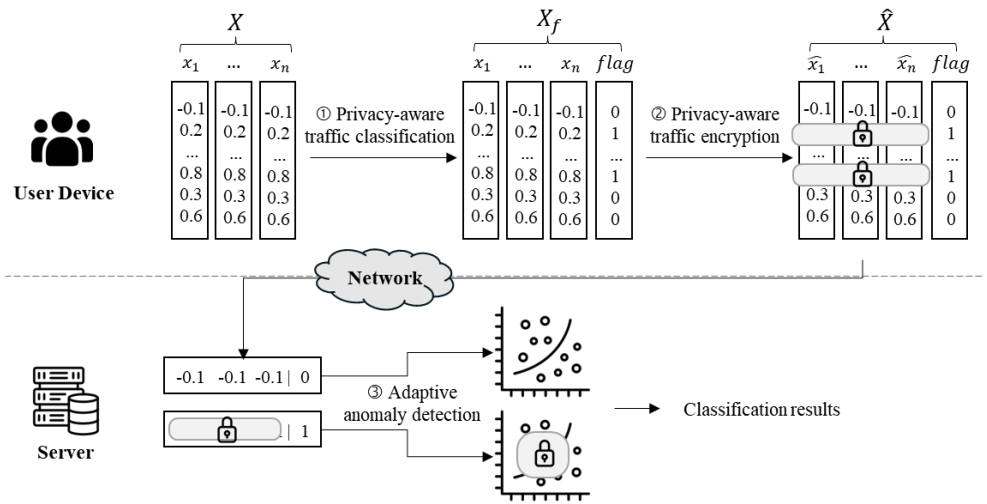
04 Conclusion and future work

Conclusions and discussions

- The results show that the proposed model reduces latency by 8.6 times compared to conventional privacy-preserving anomaly detection models, and similar information leakage to privacy-preserving anomaly detection models
- However, this study does not fully specify the mechanism for real-time classification of sensitive and non-sensitive traffic.

Future work

- Future work will explore more effective traffic-classification strategies and validate the proposed approach across a broader range of datasets.



Q&A

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